

manager,  $r$ : Giacomo works with Faheem, and  $s$ : The software application was completed on time. Translate each of the following statements to symbols, then construct a truth table to determine its truth value.

11. If Giacomo works with Faheem, then Faheem is not a software engineer.
12. If the software application was not completed on time, then Ann is not a project manager.
13. The software application was completed on time if and only if Giacomo worked with Faheem.
14. Ann is not a project manager if and only if Faheem is a software engineer.
15. If the software application was completed on time, then Ann is a project manager, but Faheem is not a software engineer.
16. If Giacomo works with Faheem and Ann is a project manager, then the software application was completed on time.
17. The software application was not completed on time if and only if Faheem is a software engineer or Giacomo did not work with Faheem.
18. Faheem is a software engineer or Ann is not a project manager if and only if Giacomo did not work with Faheem and the software application was completed on time.
19. Ann is a project manager implies Faheem is a software engineer if and only if the software application was completed on time implies Giacomo worked with Faheem.
20. If Giacomo did not work with Faheem implies that the software application was not completed on time, then Ann was not the project manager.

For the following exercises, construct a truth table to analyze all the possible outcomes and determine the validity of each argument.

21.  $p \vee \sim q \rightarrow q$
22.  $\sim q \rightarrow p \wedge \sim q$
23.  $(p \rightarrow q) \leftrightarrow q$
24.  $(p \rightarrow q) \leftrightarrow p$
25.  $\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$
26.  $(p \rightarrow q) \wedge p \rightarrow q$
27.  $p \rightarrow q \rightarrow r$
28.  $(p \rightarrow q) \wedge (q \rightarrow r) \leftrightarrow (p \rightarrow r)$
29.  $p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$
30.  $p \vee (q \vee r) \leftrightarrow (p \vee q) \vee r$

## 2.5 Equivalent Statements



**Figure 2.11** How your logical argument is stated affects the response, just like how you speak when holding a

conversation can affect how your words are received. (credit: modification of work by Goelshivi/Flickr, Public Domain Mark 1.0)

## Learning Objectives

After completing this section, you should be able to:

1. Determine whether two statements are logically equivalent using a truth table.
2. Compose the converse, inverse, and contrapositive of a conditional statement

Have you ever had a conversation with or sent a note to someone, only to have them misunderstand what you intended to convey? The way you choose to express your ideas can be as, or even more, important than what you are saying. If your goal is to convince someone that what you are saying is correct, you will not want to alienate them by choosing your words poorly.

Logical arguments can be stated in many different ways that still ultimately result in the same valid conclusion. Part of the art of constructing a persuasive argument is knowing how to arrange the facts and conclusion to elicit the desired response from the intended audience.

In this section, you will learn how to determine whether two statements are **logically equivalent** using truth tables, and then you will apply this knowledge to compose logically equivalent forms of the conditional statement. Developing this skill will provide the additional skills and knowledge needed to construct well-reasoned, persuasive arguments that can be customized to address specific audiences.

 An alternate way to think about logical equivalence is that the truth values have to match. That is, whenever  $p$  is true,  $q$  is also true, and whenever  $p$  is false,  $q$  is also false.

## Determine Logical Equivalence

Two statements,  $p$  and  $q$ , are logically equivalent when  $p \leftrightarrow q$  is a valid argument, or when the last column of the truth table consists of only true values. When a logical statement is always true, it is known as a **tautology**. To determine whether two statements  $p$  and  $q$  are logically equivalent, construct a truth table for  $p \leftrightarrow q$  and determine whether it is valid. If the last column is all true, the argument is a tautology, it is valid, and  $p$  is logically equivalent to  $q$ ; otherwise,  $p$  is not logically equivalent to  $q$ .

### EXAMPLE 2.22

#### Determining Logical Equivalence with a Truth Table

Create a truth table to determine whether the following compound statements are logically equivalent.

1.  $p \rightarrow q; \sim p \rightarrow \sim q$
2.  $p \rightarrow q; \sim p \vee q$

#### Solution

1. Construct a truth table for the biconditional formed by using the first statement as the hypothesis and the second statement as the conclusion,  $(p \rightarrow q) \leftrightarrow (\sim p \rightarrow \sim q)$ .

| $p$ | $q$ | $p \rightarrow q$ | $\sim p$ | $\sim q$ | $\sim p \rightarrow \sim q$ | $(p \rightarrow q) \leftrightarrow (\sim p \rightarrow \sim q)$ |
|-----|-----|-------------------|----------|----------|-----------------------------|-----------------------------------------------------------------|
| T   | T   | T                 | F        | F        | T                           | T                                                               |
| T   | F   | F                 | F        | T        | T                           | F                                                               |
| F   | T   | T                 | T        | F        | F                           | F                                                               |
| F   | F   | T                 | T        | T        | T                           | T                                                               |

Because the last column is not all true, the biconditional is not valid and the statement  $p \rightarrow q$  is not logically equivalent to the statement  $\sim p \rightarrow \sim q$ .

2. Construct a truth table for the biconditional formed by using the first statement as the hypothesis and the second

statement as the conclusion,  $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$ .

| $p$ | $q$ | $p \rightarrow q$ | $\sim p$ | $\sim p \vee q$ | $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$ |
|-----|-----|-------------------|----------|-----------------|-----------------------------------------------------|
| T   | T   | T                 | F        | T               | T                                                   |
| T   | F   | F                 | F        | F               | T                                                   |
| F   | T   | T                 | T        | T               | T                                                   |
| F   | F   | T                 | T        | T               | T                                                   |

Because the last column is true for every entry, the biconditional is valid and the statement  $p \rightarrow q$  is logically equivalent to the statement  $\sim p \vee q$ . Symbolically,  $p \rightarrow q \equiv \sim p \vee q$ .

### > YOUR TURN 2.22

Create a truth table to determine whether the following compound statements are logically equivalent.

1.  $p \rightarrow q$ ;  $q \rightarrow \sim p$
2.  $p \rightarrow q$ ;  $p \vee \sim q$

## Compose the Converse, Inverse, and Contrapositive of a Conditional Statement

The **converse**, **inverse**, and **contrapositive** are variations of the conditional statement,  $p \rightarrow q$ .

- The converse is if  $q$  then  $p$ , and it is formed by interchanging the hypothesis and the conclusion. The converse is logically equivalent to the inverse.
- The inverse is if  $\sim p$  then  $\sim q$ , and it is formed by negating both the hypothesis and the conclusion. The inverse is logically equivalent to the converse.
- The contrapositive is if  $\sim q$  then  $\sim p$ , and it is formed by interchanging and negating both the hypothesis and the conclusion. The contrapositive is logically equivalent to the conditional.

The table below shows how these variations are presented symbolically.

| Conditional |     |          |          | Contrapositive    | Converse                    | Inverse           |                             |
|-------------|-----|----------|----------|-------------------|-----------------------------|-------------------|-----------------------------|
| $p$         | $q$ | $\sim p$ | $\sim q$ | $p \rightarrow q$ | $\sim q \rightarrow \sim p$ | $q \rightarrow p$ | $\sim p \rightarrow \sim q$ |
| T           | T   | F        | F        | T                 | T                           | T                 | T                           |
| T           | F   | F        | T        | F                 | F                           | T                 | T                           |
| F           | T   | T        | F        | T                 | T                           | F                 | F                           |
| F           | F   | T        | T        | T                 | T                           | T                 | T                           |

### EXAMPLE 2.23

#### Writing the Converse, Inverse, and Contrapositive of a Conditional Statement

Use the statements,  $p$ : Harry is a wizard and  $q$ : Hermione is a witch, to write the following statements:

1. Write the conditional statement,  $p \rightarrow q$ , in words.
2. Write the converse statement,  $q \rightarrow p$ , in words.

3. Write the inverse statement,  $\sim p \rightarrow \sim q$ , in words.
4. Write the contrapositive statement,  $\sim q \rightarrow \sim p$ , in words.

 **Solution**

1. The conditional statement takes the form, "if  $p$ , then  $q$ ," so the conditional statement is: "If Harry is a wizard, then Hermione is a witch." Remember the *if... then...* words are the connectives that form the conditional statement.
2. The converse swaps or interchanges the hypothesis,  $p$ , with the conclusion,  $q$ . It has the form, "if  $q$ , then  $p$ ." So, the converse is: "If Hermione is a witch, then Harry is a wizard."
3. To construct the inverse of a statement, negate both the hypothesis and the conclusion. The inverse has the form, "if  $\sim p$ , then  $\sim q$ ," so the inverse is: "If Harry is not a wizard, then Hermione is not a witch."
4. The contrapositive is formed by negating and interchanging both the hypothesis and conclusion. It has the form, "if  $\sim q$ , then  $\sim p$ ," so the contrapositive statement is: "If Hermione is not a witch, then Harry is not a wizard."

 **YOUR TURN 2.23**

Use the statements,  $p$ : Elvis Presley wore capes and  $q$ : Some superheroes wear capes, to write the following statements:

1. Write the conditional statement,  $p \rightarrow q$ , in words.
2. Write the converse statement,  $q \rightarrow p$ , in words.
3. Write the inverse statement,  $\sim p \rightarrow \sim q$ , in words.
4. Write the contrapositive statement,  $\sim q \rightarrow \sim p$ , in words.

**EXAMPLE 2.24**

**Identifying the Converse, Inverse, and Contrapositive**

Use the conditional statement, "If all dogs bark, then Lassie likes to bark," to identify the following.

1. Write the hypothesis of the conditional statement and label it with a  $p$ .
2. Write the conclusion of the conditional statement and label it with a  $q$ .
3. Identify the following statement as the converse, inverse, or contrapositive: "If Lassie likes to bark, then all dogs bark."
4. Identify the following statement as the converse, inverse, or contrapositive: "If Lassie does not like to bark, then some dogs do not bark."
5. Which statement is logically equivalent to the conditional statement?

 **Solution**

1. The hypothesis is the phrase following the *if*. The answer is  $p$ : All dogs bark. Notice, the word *if* is not included as part of the hypothesis.
2. The conclusion of a conditional statement is the phrase following the *then*. The word *then* is not included when stating the conclusion. The answer is:  $q$ : Lassie likes to bark.
3. "Lassie likes to bark" is  $q$  and "All dogs bark" is  $p$ . So, "If Lassie likes to bark, then all dogs bark," has the form "if  $q$ , then  $p$ ," which is the form of the converse.
4. "Lassie does not like to bark" is  $\sim q$  and "Some dogs do not bark" is  $\sim p$ . The statement, "If Lassie does not like to bark, then some dogs do not bark," has the form "if  $\sim q$ , then  $\sim p$ ," which is the form of the contrapositive.
5. The contrapositive  $\sim q \rightarrow \sim p$  is logically equivalent to the conditional statement  $p \rightarrow q$ .

 **YOUR TURN 2.24**

Use the conditional statement, "If Dora is an explorer, then Boots is a monkey," to identify the following:

1. Write the hypothesis of the conditional statement and label it with a  $p$ .
2. Write the conclusion of the conditional statement and label it with a  $q$ .
3. Identify the following statement as the converse, inverse, or contrapositive: "If Dora is not an explorer, then Boots is not a monkey."

- Identify the following statement as the converse, inverse, or contrapositive: "If Boots is a monkey, then Dora is an explorer."
- Which statement is logically equivalent to the inverse?

**EXAMPLE 2.25****Determining the Truth Value of the Converse, Inverse, and Contrapositive**

Assume the conditional statement,  $p \rightarrow q$ : "If Chadwick Boseman was an actor, then Chadwick Boseman did not star in the movie *Black Panther*" is false, and use it to answer the following questions.

- Write the converse of the statement in words and determine its truth value.
- Write the inverse of the statement in words and determine its truth value.
- Write the contrapositive of the statement in words and determine its truth value.

 **Solution**

- The only way the conditional statement can be false is if the hypothesis,  $p$ : Chadwick Boseman was an actor, is true and the conclusion,  $q$ : Chadwick Boseman did not star in the movie *Black Panther*, is false. The converse is  $q \rightarrow p$ , and it is written in words as: "If Chadwick Boseman did not star in the movie *Black Panther*, then Chadwick Boseman was an actor." This statement is true, because false  $\rightarrow$  true is true.
- The inverse has the form " $\sim p \rightarrow \sim q$ ." The written form is: "If Chadwick Boseman was not an actor, then Chadwick Boseman starred in the movie *Black Panther*." Because  $p$  is true, and  $q$  is false,  $\sim p$  is false, and  $\sim q$  is true. This means the inverse is false  $\rightarrow$  true, which is true. Alternatively, from Question 1, the converse is true, and because the inverse is logically equivalent to the converse it must also be true.
- The contrapositive is logically equivalent to the conditional. Because the conditional is false, the contrapositive is also false. This can be confirmed by looking at the truth values of the contrapositive statement. The contrapositive has the form " $\sim q \rightarrow \sim p$ ." Because  $q$  is false and  $p$  is true,  $\sim q$  is true and  $\sim p$  is false. Therefore,  $\sim q \rightarrow \sim p$  is true  $\rightarrow$  false, which is false. The written form of the contrapositive is "If Chadwick Boseman starred in the movie *Black Panther*, then Chadwick Boseman was not an actor."

 **YOUR TURN 2.25**

Assume the conditional statement  $p \rightarrow q$ : "If my friend lives in San Francisco, then my friend does not live in California" is false, and use it to answer the following questions.

- Write the converse of the statement in words and determine its truth value.
- Write the inverse of the statement in words and determine its truth value.
- Write the contrapositive of the statement in words and determine its truth value.

**Check Your Understanding**

- Two statements  $p$  and  $q$  are logically equivalent to each other if the biconditional statement,  $p \leftrightarrow q$  is \_\_\_\_\_.
- The \_\_\_\_\_ statement has the form, " $p$  then  $q$ ."
- The contrapositive is \_\_\_\_\_ to the conditional statement, and has the form, "if  $\sim q$ , then  $\sim p$ ."
- The \_\_\_\_\_ of the conditional statement has the form, "if  $\sim p$ , then  $\sim q$ ."
- The \_\_\_\_\_ of the conditional statement is logically equivalent to the \_\_\_\_\_ and has the form, "if  $q$  then  $p$ ."

**SECTION 2.5 EXERCISES**

For the following exercises, determine whether the pair of compound statements are logically equivalent by constructing a truth table.

- Converse:  $q \rightarrow p$  and inverse:  $\sim p \rightarrow \sim q$
- Conditional:  $p \rightarrow q$  and contrapositive:  $\sim q \rightarrow \sim p$

3. Inverse:  $\sim p \rightarrow \sim q$  and contrapositive:  $\sim q \rightarrow \sim p$
4. Conditional:  $p \rightarrow q$  and converse:  $q \rightarrow p$
5.  $\sim p \rightarrow q$  and  $p \vee \sim q$
6.  $\sim p \rightarrow q$  and  $p \vee q$
7.  $\sim(p \wedge q)$  and  $\sim p \wedge \sim q$
8.  $\sim(p \wedge q)$  and  $\sim p \vee \sim q$
9.  $p \wedge (q \vee r)$  and  $(p \wedge q) \vee (p \wedge r)$
10.  $p \wedge (q \vee r)$  and  $(p \wedge q) \vee r$

For the following exercises, answer the following:

- a. Write the conditional statement  $p \rightarrow q$  in words.
  - b. Write the converse statement  $q \rightarrow p$  in words.
  - c. Write the inverse statement  $\sim p \rightarrow \sim q$  in words.
  - d. Write the contrapositive statement  $\sim q \rightarrow \sim p$  in words.
11.  $p$ : Six is afraid of Seven and  $q$ : Seven ate Nine.
  12.  $p$ : Hope is eternal and  $q$ : Despair is temporary.
  13.  $p$ : Tom Brady is a quarterback and  $q$ : Tom Brady does not play soccer.
  14.  $p$ : Shakira does not sing opera and  $q$ : Shakira sings popular music.
  15.  $p$ : The shape does not have three sides and  $q$ : The shape is not a triangle.
  16.  $p$ : All birds can fly and  $q$ : Emus can fly.
  17.  $p$ : Penguins cannot fly and  $q$ : Some birds can fly.
  18.  $p$ : Some superheroes do not wear capes and  $q$ : Spiderman is a superhero.
  19.  $p$ : No Pokémon are little ponies and  $q$ : Bulbasaur is a Pokémon.
  20.  $p$ : Roses are red, and violets are blue and  $q$ : Sugar is sweet, and you are sweet too.

For the following exercises, use the conditional statement: "If Clark Kent is Superman, then Lois Lane is not a reporter," to answer the following questions.

21. Write the hypothesis of the conditional statement, label it with a  $p$ , and determine its truth value.
22. Write the conclusion of the conditional statement, label it with a  $q$ , and determine its truth value.
23. Identify the following statement as the converse, inverse, or contrapositive, and determine its truth value: "If Clark Kent is not Superman, then Lois Lane is a reporter."
24. Identify the following statement as the converse, inverse, or contrapositive, and determine its truth value: "If Lois Lane is a reporter, then Clark Kent is not Superman."
25. Which form of the conditional is logically equivalent to the converse?

For the following exercises, use the conditional statement: "If *The Masked Singer* is not a music competition, then Donnie Wahlberg was a member of New Kids on the Block," to answer the following questions.

26. Write the hypothesis of the conditional statement, label it with a  $p$ , and determine its truth value.
27. Write the conclusion of the conditional statement, label it with a  $q$ , and determine its truth value.
28. Identify the following statement as the converse, inverse, or contrapositive, and determine its truth value: "If Donnie Wahlberg was a member of New Kids on the Block, then *The Masked Singer* is not a music competition."
29. Identify the following statement as the converse, inverse, or contrapositive, and determine its truth value: "If *The Masked Singer* is a music competition, then Donnie Wahlberg was not a member of New Kids on the Block."
30. Which form of the conditional is logically equivalent to the contrapositive,  $\sim q \rightarrow \sim p$ ?

For the following exercises, use the conditional statement: "If all whales are mammals, then no fish are whales," to answer the following questions.

31. Write the hypothesis of the conditional statement, label it with a  $p$ , and determine its truth value.
32. Write the conclusion of the conditional statement, label it with a  $q$ , and determine its truth value.
33. Identify the following statement as the converse, inverse, or contrapositive, and determine its truth value: "If some fish are whales, then some whales are not mammals."
34. Write the inverse in words and determine its truth value.
35. Write the converse in words and determine its truth value.

For the following exercises, use the conditional statement: "If some parallelograms are rectangles, then some circles are not symmetrical," to answer the following questions.

36. Write the hypothesis of the conditional statement, label it with a  $p$ , and determine its truth value.
37. Write the conclusion of the conditional statement, label it with a  $q$ , and determine its truth value.
38. Write the converse in words and determine its truth value.
39. Write the contrapositive in words and determine its truth value.